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Lab Experiment 7

AIM

To design and implement simple views, complex views, and materialized views, and analyze their performance differences to understand their impact on query optimization.

Objective of the Session

- To create simple and complex views
- To implement materialized views
- To compare execution performance of different views
- To understand how materialized views improve query efficiency.

Software Requirements

Database Management System:

Oracle Database Express Edition (Oracle XE)

PostgreSQL Database

Database Administration Tool / Client Tool:

Oracle SQL Developer (for Oracle XE)

pgAdmin (for PostgreSQL)

Experiment Question :

Create:

1. A simple view based on a single table
2. A complex view using joins and aggregation
3. A materialized view storing precomputed results

Then compare their performance using execution time.

Practical/Experiment steps:

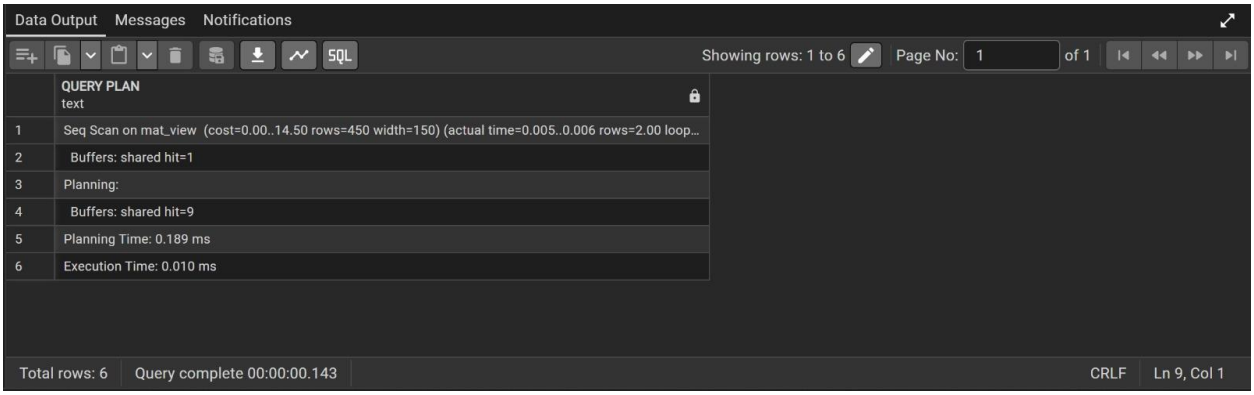
1. Create employee and department tables
2. Insert sample data
3. Create a simple view
4. Create a complex view using joins and aggregation
5. Create a materialized view
6. Run SELECT queries on all views
7. Compare execution time using EXPLAIN ANALYZE
8. Observe and analyze performance differences

CODE:

```
-- Create Tables
CREATE TABLE department (
   dept_id INT,
   dept_name VARCHAR(50)
);
```

```sql
CREATE TABLE employee (
    emp_id INT,
    emp_name VARCHAR(50),
    salary INT,
    dept_id INT
);

-- Insert Data
INSERT INTO department VALUES (1, 'HR'), (2, 'IT');

INSERT INTO employee VALUES
(1, 'Simon', 30000, 1),
(2, 'Alex', 50000, 2),
(3, 'John', 25000, 1);

-- Simple View
CREATE VIEW simple_view AS
SELECT emp_id, emp_name, salary FROM employee;

-- Complex View (Join + Aggregation)
CREATE VIEW complex_view AS
SELECT d.dept_name, AVG(e.salary) AS avg_salary
FROM employee e
JOIN department d ON e.dept_id = d.dept_id
GROUP BY d.dept_name;

-- Materialized View (PostgreSQL)
CREATE MATERIALIZED VIEW mat_view AS
SELECT d.dept_name, AVG(e.salary) AS avg_salary
FROM employee e
JOIN department d ON e.dept_id = d.dept_id
GROUP BY d.dept_name;

-- Performance Check
EXPLAIN ANALYZE SELECT * FROM simple_view;
EXPLAIN ANALYZE SELECT * FROM complex_view;
EXPLAIN ANALYZE SELECT * FROM mat_view;
```

OUTPUT:



Learning Outcome

After completing this experiment, the student will be able to:

- Understand the concept of simple, complex, and materialized views
- Differentiate between dynamic views and stored results
- Analyze query performance using execution time
- Understand how materialized views improve efficiency
- Apply optimization techniques in real-world database systems